Thompson's group F is nonamenable

OpenAI

September 23, 2026

Abstract

We prove that Thompson's group F is nonamenable. This confirms Geoghegan's conjecture and resolves the amenability problem for F.

1 Introduction

Thompson's group F consists of the increasing piecewise linear homeomorphisms of $[0,1]$ with finitely many pieces, dyadic rational breakpoints, and slopes in $2^{\mathbb{Z}}$. We use the multiplication convention $hg = h \circ g$ and regard F as a discrete group.

A discrete group G is *amenable* if its bounded real-valued functions admit a positive normalized left-invariant mean: a linear functional $M : \ell^\infty(G;\mathbb{R}) \to \mathbb{R}$ satisfying $M(1) = 1$, $M(\varphi) \geq 0$ whenever $\varphi \geq 0$, and $M(g \mapsto \varphi(hg)) = M(\varphi)$ for every $h \in G$. This invariant-mean viewpoint belongs to the classical theory developed by von Neumann and Day [19, 6]. We use the following forward implication of the Følner criterion [7]: if G is amenable, then for every finite $S \subset G$ and every $\varepsilon > 0$, there is a nonempty finite $A \subset G$ such that

$$\frac{|hA \triangle A|}{|A|} < \varepsilon \qquad (h \in S). \tag{1.1}$$

The modern left-translation formulation in [14, Theorem 1.2] bounds the sum of these ratios and hence implies each strict individual bound by taking a smaller tolerance. Thus amenability requires finite sets whose relative boundaries are simultaneously small under any fixed finite family of translations.

Thompson introduced F in 1965; an early published construction appears in McKenzie and Thompson [11], with the identification explained by Cannon, Floyd, and Parry [5, pp. 215–216]. The latter authors record Geoghegan's 1979 conjecture that F is nonamenable [5, p. 227]. We prove this conjecture.

Theorem 1.1. *Thompson's group F is not amenable.*

Context and prior work. Two classical structural facts help explain the difficulty of the amenability problem. Brin and Squier show that F contains no nonabelian free subgroup [3, Theorem 3.1], so the familiar free-subgroup obstruction does not apply. Yet F is not elementary amenable [5, Theorem 4.10]: it lies outside the class generated by finite and abelian groups under subgroups, quotients, extensions, and directed unions. Neither fact decides ordinary amenability. The group also has strong topological finiteness properties: Brown and Geoghegan construct a classifying space with finitely many cells in each dimension [4].

Finite approximations have provided several ways to study the remaining question. Moore proves tower lower bounds on the sizes of possible Følner sets [15, Theorem 1.1] and characterizes amenability of F by a scalar convex Ramsey property for finite rooted ordered binary trees [14, Theorem 3.1]. Guba improves densities of finite Cayley subgraphs [8] and derives restrictions on right-invariant means from a partition into seven diagram-defined classes [9]. From an operator-algebraic direction, Haagerup and Olesen prove that simplicity of the reduced group C^*-algebra of Thompson's group T would imply nonamenability of F [10, Theorem 5.5]. These results expose constraints on amenability through finite sets, trees, diagrams, and representations. The proof below uses finite dyadic partitions and scalar correlations of Hilbert-valued colors.

There have also been claims of a complete resolution in both directions. Akhmedov's 2021 version claims nonamenability through a height-function criterion [1]. Shavgulidze claimed amenability [18]; Moore identified errors in that approach [12]. Moore separately withdrew an amenability claim after Akhmedov identified an error in Lemma 4.13 of that manuscript [13]. That withdrawal concerns a different work from his published Ramsey characterization cited above.

Proof strategy. The proof turns approximate translation invariance of finite averages of scalar functions on F into an approximate fixed point of a Lipschitz map on a Hilbert ball. The following infinite-dimensional phenomenon supplies a map for which that conclusion is impossible.

Lemma 1.2 (Benyamini–Sternfeld)**.** *There are a real Hilbert space H, a Lipschitz map $f : B \to B$ on its closed unit ball, and a constant $\delta > 0$ such that*

$$\|f(x) - x\| \geq \delta \qquad (x \in B).$$

Benyamini and Sternfeld prove this assertion for every infinite-dimensional normed space [2, Theorem, part (3), p. 439]. For completeness, Appendix A constructs such a map on $L^2([0,1];\mathbb{R}^2)$ with $\delta = 1/2$, including all the needed Lipschitz estimates.

Fix this map f, a positive Lipschitz constant L, and a large integer D. A basic dyadic interval is a cell of a uniform dyadic partition, and a basic partition is a finite partition into such cells. Choose D basic intervals with positive gaps between them and endpoints in $(0,1)$, called parents. Put an affine copy of this family inside each parent, calling the copies descendants. When a parent is a union of partition cells, restricting to it and stretching back to $[0,1]$ gives its normalized restriction. Color each basic partition recursively by applying f to the mean of its parent-restriction colors when all these restrictions are defined, and by zero otherwise. Proper restriction decreases the number of cells, so the recursion is well founded. For a sufficiently fine image partition under $g \in F$, its parent colors X_i and descendant colors Y_{ij} therefore satisfy

$$m = \frac{1}{D}\sum_{i=1}^{D} X_i, \qquad z_i = \frac{1}{D}\sum_{j=1}^{D} Y_{ij}, \qquad X_i = f(z_i) \quad (1 \leq i \leq D).$$

The group F can carry any ordered pair of separated basic intervals with endpoints in $(0,1)$ affinely onto any other such pair. Choose a finite set S of these transports comparing every separated pair among the parents and descendants with one reference pair. The set S is fixed before the arbitrary nonempty finite set $A \subset F$; only the common partition level, chosen fine enough for $A \cup SA$, may depend on A. Write η for the largest relative S-boundary of A. Exact covariance of normalized restrictions and finite-average cancellation make all averaged scalar correlations of separated interval colors differ from one common value by at most η.

Expand the average over A of $\|z_i - m\|^2$. The common correlation cancels; diagonal and nested pairs occupy only a fraction $1/D$ of each relevant sum, as Figure 1 illustrates. Unit-ball bounds

control those exceptions, giving a universal constant times $D^{-1}+\eta$. Since m is also the mean of the $f(z_i)$, convexity and Lipschitz continuity bound the average squared displacement $\|m-f(m)\|^2$ by a constant times $L^2(D^{-1}+\eta)$. For large D and small η, this contradicts the uniform lower bound δ^2. Section 2 makes the argument precise and proves a uniform positive boundary bound in Proposition 2.3, contradicting (1.1).

Section 3 records consequences for uniformly bounded representations and percolation, using separate companion theorems.

2 Recursive colors and finite averages

We prove Theorem 1.1 using the map supplied by Lemma 1.2. The argument uses a single finite average over the group. We first record the dyadic facts that make its correlations comparable.

2.1 Dyadic partitions and affine transport

We use the standard dyadic-partition model of F; see [5, Section 2, Lemma 2.2]. We record the restriction and pair-transport facts needed for the finite averages.

A *basic dyadic interval* is an interval

$$[k2^{-r},(k+1)2^{-r}]\subseteq[0,1],\qquad r\in\mathbb{N}\cup\{0\},\quad 0\le k<2^r.$$

A *basic partition* is a finite partition of $[0,1]$ into such intervals; cells may share endpoints. Its mesh is the greatest length of a cell. A partition T *respects* an interval I if I is a union of cells of T.

For a basic interval I, let $s_I:[0,1]\to I$ be the increasing affine map, and write $I\cdot J=s_I(J)$. If T respects I, define its normalized restriction by

$$T_I=\{s_I^{-1}(K):K\in T,\ K\subseteq I\}.$$

Both $I\cdot J$ and the cells of T_I are basic dyadic intervals. Indeed, if I has length 2^{-r}, a basic cell contained in I has length 2^{-s} with $s\ge r$; applying s_I^{-1} gives a cell of length $2^{-(s-r)}$ whose left endpoint is a multiple of that length. The affine charts satisfy $s_{I\cdot J}=s_I\circ s_J$. Consequently, whenever T respects both I and $I\cdot J$, the partition T_I respects J and

$$(T_I)_J=T_{I\cdot J}. \tag{2.1}$$

Let $T^{(n)}$ be the uniform partition of $[0,1]$ into intervals of length 2^{-n}. For $g\in F$, the notation $gT^{(n)}$ denotes the partition formed by the images of these cells.

Lemma 2.1. *For each $g\in F$, the partitions $gT^{(n)}$ are basic for all sufficiently large n, and their meshes tend to zero. In particular, given a finite set $K\subset F$ and a finite family $\mathcal{J}$ of basic intervals, there is an n_0 such that $gT^{(n)}$ is basic and respects every member of $\mathcal{J}$ whenever $g\in K$ and $n\ge n_0$.*

Proof. The image under g of each breakpoint is dyadic: starting from $g(0)=0$, sum the increments on the preceding linear pieces, each a power of two times a dyadic length. Thus every linear piece has the form $x\mapsto 2^qx+b$ with $q\in\mathbb{Z}$ and b dyadic. Choose n large enough to resolve all breakpoints, to have $n\ge q$ on every piece, and to make each corresponding b an integer multiple of 2^{q-n}. The image of each uniform cell is then a basic interval of length 2^{q-n}. There are finitely many slopes, so the mesh tends to zero.

Two basic intervals with intersecting interiors are nested. Hence a basic partition of mesh smaller than the length of a basic interval I respects I: every cell whose interior meets that of I must be contained in I. Apply this observation to the finitely many intervals in $\mathcal{J}$, and then take the maximum of the finitely many thresholds for $g\in K$. □

Write $I < J$ when $\max I < \min J$; in particular this notation requires a positive gap. An interval is *internal* if both its endpoints lie in $(0, 1)$.

Lemma 2.2. *If $I < J$ and $I' < J'$ are pairs of internal basic intervals, there is an $h \in F$ whose restrictions carry I affinely onto I' and J affinely onto J'.*

Proof. The two selected intervals leave three complementary gaps, all of positive length and with dyadic endpoints. Partition each gap into basic cells, for example by a sufficiently fine uniform dyadic grid. Do this for both pairs. Within each corresponding pair of gaps, equalize the numbers of cells by successively bisecting cells on the side with fewer cells. Each bisection increases the number by one. The source and target partitions now have the same number of cells, with the two selected intervals in matching positions. Map corresponding cells increasingly and affinely. All breakpoints are dyadic and all slopes are ratios of basic lengths, hence powers of two. This defines the required element of F. □

The relevant covariance is exact. If h carries I affinely onto I', then $h \circ s_I = s_{I'}$. Thus, if $gT^{(n)}$ and $(hg)T^{(n)}$ are basic and respect I and I', respectively,

$$\big((hg)T^{(n)}\big)_{I'} = (gT^{(n)})_I. \tag{2.2}$$

Here, as throughout, $hg = h \circ g$. To verify the equality, match each source cell C with its image $h(C)$ and apply $s_{I'}^{-1} \circ h = s_I^{-1}$ on I.

2.2 The recursive coloring

Fix a real Hilbert space H, its closed unit ball B, and a Lipschitz map $f : B \to B$ as in Lemma 1.2. Fix $L > 0$ and $\delta > 0$ such that

$$\|f(x) - f(y)\| \le L\|x - y\|, \qquad \|f(x) - x\| \ge \delta \quad (x, y \in B). \tag{2.3}$$

Choose an integer $D \ge 2$ so that

$$D > \frac{4L^2}{\delta^2}. \tag{2.4}$$

Choose internal basic intervals $I_1 < \cdots < I_D$. Such a family exists for every finite D: take $2^r > 2D$ and use $I_j = [(2j-1)2^{-r}, 2j2^{-r}]$.

Define a color $p(T) \in B$ for every basic partition T by induction on its number of cells:

$$p(T) = \begin{cases} f\left(\dfrac{1}{D}\displaystyle\sum_{j=1}^{D} p(T_{I_j})\right), & \text{if } T \text{ respects every } I_j, \\ 0, & \text{otherwise.} \end{cases} \tag{2.5}$$

This induction is well founded. In the first case each T_{I_j} has strictly fewer cells than T, since I_j is internal and T has cells outside it. The average of the previously defined colors belongs to the convex ball B, so the argument of f is always in its domain, and $p(T) \in B$.

For the displayed choice $|I_j| = 2^{-r}$, the one-cell partition $T^{(0)}$ has color zero, while

$$p(T^{(r)}) = f(0), \qquad p(T^{(2r)}) = f(f(0)).$$

Indeed, normalizing a restriction to any I_j changes these uniform partition levels from r to 0 and from $2r$ to r, respectively.

We use the finite family of parent intervals and their descendants

$$\mathcal{I} = \{I_i : 1 \le i \le D\} \cup \{I_i \cdot I_j : 1 \le i, j \le D\}.$$

All these intervals are internal. Call (n, g) *admissible* if $gT^{(n)}$ is a basic partition respecting every member of $\mathcal{I}$. For every n, every $g \in F$, and every $I \in \mathcal{I}$, define

$$X_I^{(n)}(g) = \begin{cases} p\big((gT^{(n)})_I\big), & \text{if } (n, g) \text{ is admissible,} \\ 0, & \text{otherwise.} \end{cases} \tag{2.6}$$

In particular these are globally defined B-valued functions. Their scalar correlations

$$\varphi_{I,J}^{(n)}(g) = \langle X_I^{(n)}(g), X_J^{(n)}(g) \rangle$$

are defined on all of F and have absolute value at most one. By Lemma 2.1, every finite set of group elements is admissible at one common sufficiently large level n.

2.3 A uniform boundary bound

For every ordered pair $I < J$ in $\mathcal{I}$, choose, using Lemma 2.2, an element $h_{I,J} \in F$ carrying I, J affinely onto I_1, I_2, respectively. Let S be the finite set of these elements. All choices so far, in particular S, are independent of the finite set to be tested.

For any nonempty finite $A \subset F$, any $h \in F$, and any $\varphi : F \to [-1, 1]$, cancellation over $hA \cap A$ gives

$$\left| \frac{1}{|A|} \sum_{g \in A} \varphi(hg) - \frac{1}{|A|} \sum_{g \in A} \varphi(g) \right| \le \frac{|hA \triangle A|}{|A|}. \tag{2.7}$$

This is the finite-average comparison we will apply to the scalar correlations.

Proposition 2.3. *For every nonempty finite set $A \subset F$,*

$$\max_{h \in S} \frac{|hA \triangle A|}{|A|} \ge \frac{\delta^2/L^2 - 4/D}{4(1 - 1/D)} > 0. \tag{2.8}$$

Proof. Fix such a set A and put

$$\eta = \max_{h \in S} \frac{|hA \triangle A|}{|A|}. \tag{2.9}$$

Choose a single n for which (n, g) is admissible for all

$$g \in A \cup SA, \qquad SA = \bigcup_{h \in S} hA.$$

This choice is possible because that union is finite; the level n may depend on A. We now fix this n, omit its superscript, and write

$$\mathbb{E}_A \psi = \frac{1}{|A|} \sum_{g \in A} \psi(g), \qquad \alpha = \mathbb{E}_A \varphi_{I_1, I_2}.$$

Comparing correlations. For $I < J$ in $\mathcal{I}$ and $g \in A$, admissibility of both g and $h_{I,J}g$, together with (2.2), gives

$$X_I(g) = X_{I_1}(h_{I,J}g), \qquad X_J(g) = X_{I_2}(h_{I,J}g).$$

It follows that $\varphi_{I,J}(g) = \varphi_{I_1,I_2}(h_{I,J}g)$. Apply (2.7) to the globally bounded function φ_{I_1,I_2} and use (2.9). We obtain

$$|\mathbb{E}_A\langle X_I, X_J\rangle - \alpha| \le \eta \qquad \text{whenever } I, J \in \mathcal{I} \text{ are strictly separated.} \tag{2.10}$$

The reversed order follows from symmetry of the real inner product. No condition is imposed on nested pairs. Also, a transport $h_{I,J}$ need not preserve the other intervals in $\mathcal{I}$: admissibility was required separately at its source and target group elements.

The variance estimate. For $g \in A$, put

$$m(g) = \frac{1}{D}\sum_{k=1}^{D} X_{I_k}(g), \qquad z_i(g) = \frac{1}{D}\sum_{j=1}^{D} X_{I_i \cdot I_j}(g) \quad (1 \le i \le D).$$

These vectors lie in B. Admissibility ensures that the partition $(gT^{(n)})_{I_i}$ respects each I_j. Applying the recursion and (2.1) therefore yields the exact identities

$$X_{I_i}(g) = f(z_i(g)) \qquad (g \in A, 1 \le i \le D). \tag{2.11}$$

Consequently, for every $g \in A$, the displacement bound, the convexity of the squared norm, and the Lipschitz bound give

$$\begin{aligned}
\delta^2 &\le \|m(g) - f(m(g))\|^2 \\
&= \left\| \frac{1}{D}\sum_{i=1}^{D} (f(z_i(g)) - f(m(g))) \right\|^2 \\
&\le \frac{1}{D}\sum_{i=1}^{D} \|f(z_i(g)) - f(m(g))\|^2 \\
&\le \frac{L^2}{D}\sum_{i=1}^{D} \|z_i(g) - m(g)\|^2.
\end{aligned} \tag{2.12}$$

We bound the finite average of each squared distance in the last line. In the expansion of $\|m\|^2$, the $D(D-1)$ off-diagonal terms pair strictly separated parents, so (2.10) applies. Each of the D diagonal terms is at most one. The same statements hold for the siblings in $\|z_i\|^2$. Hence

$$\mathbb{E}_A\|m\|^2 \le \left(1 - \frac{1}{D}\right)(\alpha + \eta) + \frac{1}{D}, \tag{2.13}$$

$$\mathbb{E}_A\|z_i\|^2 \le \left(1 - \frac{1}{D}\right)(\alpha + \eta) + \frac{1}{D}. \tag{2.14}$$

In the mixed term

$$\langle z_i, m\rangle = \frac{1}{D^2}\sum_{j=1}^{D}\sum_{k=1}^{D} \langle X_{I_i \cdot I_j}, X_{I_k}\rangle,$$

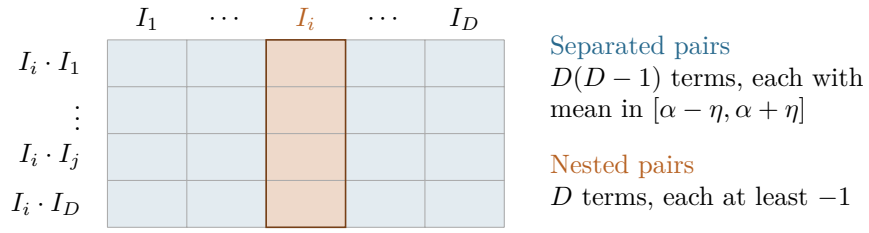


Figure 1: For fixed i, the (j,k) entry is $\langle X_{I_i \cdot I_j}, X_{I_k} \rangle$. Rows index the children of I_i and columns index all parents; the grid is schematic. Only column i contains nested pairs, a fraction $1/D$ of the D^2 terms.

the $D(D-1)$ terms with $k \neq i$ pair strictly separated intervals. Each has average at least $\alpha - \eta$. The remaining D terms pair a descendant with its own parent; each is at least -1 by the unit-ball bound. Figure 1 shows this division of the mixed pairs. Thus

$$\mathbb{E}_A \langle z_i, m \rangle \geq \left(1 - \frac{1}{D}\right)(\alpha - \eta) - \frac{1}{D}. \tag{2.15}$$

Expanding the squared distance and using (2.13)–(2.15), the coefficients of α cancel and give

$$\mathbb{E}_A \|z_i - m\|^2 \leq \frac{4}{D} + 4\left(1 - \frac{1}{D}\right)\eta \qquad (1 \leq i \leq D). \tag{2.16}$$

This estimate does not require any sign assumption on α.

Average (2.12) over A and apply (2.16):

$$\begin{aligned} \delta^2 &\leq \frac{L^2}{D} \sum_{i=1}^{D} \mathbb{E}_A \|z_i - m\|^2 \\ &\leq L^2 \left(\frac{4}{D} + 4\left(1 - \frac{1}{D}\right)\eta\right). \end{aligned}$$

Rearranging proves (2.8); its right-hand side is positive by (2.4) and does not depend on A. □

Proof of Theorem 1.1. The fixed finite set S and the uniform positive lower bound in Proposition 2.3 violate the Følner criterion (1.1). Hence F is not amenable. □

3 Consequences

Theorem 1.1 allows us to apply two companion results for nonamenable groups. The first concerns representations that are uniformly close to being isometric but cannot be made unitary by an equivalent Hilbert norm. The second concerns percolation on the simple Cayley graphs of F associated with finite symmetric generating sets. Neither companion result is used in the proof of nonamenability.

3.1 Uniformly bounded representations

Corollary 3.1. *For every $\varepsilon > 0$, there are a separable complex Hilbert space H and a representation $\pi : F \to \mathrm{GL}(H)$ such that $\sup_{g \in F} \|\pi(g)\| \leq 1 + \varepsilon$, but no bounded invertible operator S on H makes every $S\pi(g)S^{-1}$ unitary.*

Proof. The group F is countable, by its finite dyadic piecewise linear description, and nonamenable by Theorem 1.1, so the conclusion follows from the companion unitarizability theorem [17, Theorem 1.1]. □

3.2 Percolation on Cayley graphs

For a finite symmetric generating set $S \subset F \setminus \{\mathrm{id}\}$, let G_S be the simple undirected Cayley graph with edges $\{x, xs\}$. In Bernoulli bond percolation on this fixed graph, each edge is independently open with probability p. Write p_c for the threshold for existence of an infinite open cluster and p_u for the threshold for almost-sure uniqueness of the infinite open cluster. The connection kernel $T_p(x, y) = \mathbb{P}_p(x \leftrightarrow y)$ acts on counting-measure $\ell^2(F)$, initially on finitely supported functions by summation over y; put $p_{2\to 2} = \sup\{p \in [0, 1] : T_p \text{ is bounded on } \ell^2(F)\}$.

Corollary 3.2. *For every such generating set S,*

$$\|T_{p_c}\|_{2\to 2} < \infty, \qquad p_c < p_{2\to 2} \le p_u.$$

For each fixed $p \in (p_c, p_u)$, there are almost surely infinitely many infinite open clusters. Moreover, there are deterministic $p_c < p_1 < p_2 < 1$, depending on S, such that in the coupling with independent uniform $[0, 1]$ edge labels U_e, opening e when $U_e \le p$, there are almost surely infinitely many infinite open clusters simultaneously for every $p \in [p_1, p_2]$.

Proof. Apply the companion percolation result [16, Corollary 1.2] to F, which is nonamenable by Theorem 1.1. □

A An explicit map with positive displacement

We give a direct construction of the map needed in Lemma 1.2. This proves the Hilbert-space instance of the Benyamini–Sternfeld theorem [2] directly. The aim is to construct a Lipschitz map $G : B \to H$ bounded away from zero and equal to the identity when $\|x\| \ge 1/2$. Then $f(x) = -G(x)/\|G(x)\|$ has the required displacement. A tube about a curve with a bounded tail supplies G.

Proposition A.1. *Let $H = L^2([0, 1]; \mathbb{R}^2)$ be a real Hilbert space and let $B = \{x \in H : \|x\| \le 1\}$. There is a globally Lipschitz map $f : B \to B$ such that*

$$\|f(x)\| = 1, \qquad \|f(x) - x\| \ge \frac{1}{2} \quad (x \in B).$$

Define functions on $\mathbb{R}$ by

$$a(t) = \begin{cases} t/8, & t \le 2, \\ 5/16 - (3 - t)^2/16, & 2 < t < 3, \\ 5/16, & t \ge 3, \end{cases} \qquad \theta(t) = \begin{cases} 0, & t \le 1, \\ (t - 1)^2/2, & 1 < t < 2, \\ t - 3/2, & t \ge 2. \end{cases}$$

For $w \in [0, 1]$, put

$$v(t)(w) = (\cos(\theta(t)w), \sin(\theta(t)w)), \qquad \gamma(t) = a(t)v(t).$$

Thus $\|v(t)\| = 1$, $\gamma(0) = 0$, and γ follows a straight line for $t \le 1$; its tail for $t \ge 1$ lies in the ball of radius $5/16$. The next lemma shows that, although the positive tail is bounded, curve points whose parameters differ by a fixed amount remain uniformly separated. This permits a tube of fixed radius along the whole curve.

Lemma A.2. *The curve γ is injective, has a bounded Lipschitz derivative, and*

$$c := \inf_{t\in\mathbb{R}} \|\gamma'(t)\| = \frac{1}{8}.$$

Its unit tangent $u(t) = \gamma'(t)/\|\gamma'(t)\|$ is Lipschitz, $\langle u(t), v(t)\rangle \geq 0$, and $u(t) = v(t)$ for $t \leq 1$. There is a radius $0 < \rho < 1/32$ such that every point in

$$U = \{x \in H : \operatorname{dist}(x, \gamma(\mathbb{R})) < \rho\}$$

has a unique nearest point $\gamma(t(x))$. Writing $e(x) = x - \gamma(t(x))$, we have $e(x) \perp u(t(x))$, and

$$|t(x) - t(y)| \leq 16\|x - y\| \quad \text{if } x, y \in U \text{ and } \|x - y\| < \rho. \tag{A.1}$$

Proof. Let $J(\xi_1, \xi_2) = (-\xi_2, \xi_1)$. Since $0 \leq w \leq 1$, the scalar trigonometric Taylor remainders are uniform in w, which justifies differentiation in H. We obtain

$$\gamma'(t)(w) = a'(t)v(t)(w) + a(t)\theta'(t)wJv(t)(w), \qquad \|\gamma'(t)\|^2 = a'(t)^2 + \frac{a(t)^2\theta'(t)^2}{3}.$$

For $t \leq 2$ the first term is $1/64$; for $t \geq 2$ we have $a(t) \geq 1/4$ and $\theta'(t) = 1$. Equality in the claimed lower speed bound holds for $t \leq 1$. The first derivatives match at the junctions $1, 2, 3$, and their derivatives on the intervening intervals are bounded. On the unbounded negative interval the angular derivatives vanish. Consequently γ' is bounded and Lipschitz. The same is true of v and u, and $\langle u(t), v(t)\rangle = a'(t)/\|\gamma'(t)\| \geq 0$.

We need the following uniform separation property:

$$\sigma(d) := \inf_{|t-s|\geq d} \|\gamma(t) - \gamma(s)\| > 0 \qquad (d > 0). \tag{A.2}$$

Indeed, with $\operatorname{sinc}(q) = \sin(q)/q$ for $q \neq 0$ and $\operatorname{sinc}(0) = 1$, we have

$$\langle v(s), v(t)\rangle = \operatorname{sinc}(\theta(t) - \theta(s)).$$

Since $|\operatorname{sinc}(q)| < 1$ for $q \neq 0$, distinct angular parameters give noncollinear unit vectors. On $(-\infty, 1]$ the coefficient a is strictly increasing, whereas θ is strictly increasing on $(1, \infty)$ and a is positive there. This proves injectivity.

If (A.2) failed, choose $s_n < t_n$ with $t_n - s_n \geq d$ and $\|\gamma(t_n) - \gamma(s_n)\| \to 0$, and pass to subsequences for which both parameters have limits in the extended real line. Finite limits contradict injectivity. If both limits are $-\infty$, the distances equal $(t_n - s_n)/8$ eventually. If only $s_n \to -\infty$, the first radii tend to infinity while the second remain bounded. If $s_n \to s \in \mathbb{R}$ and $t_n \to +\infty$, the displayed inner product tends to zero, and the squared distances tend to $a(s)^2 + (5/16)^2$. Finally, if both parameters tend to $+\infty$, then eventually

$$\|\gamma(t_n) - \gamma(s_n)\|^2 = 2(5/16)^2\big(1 - \operatorname{sinc}(t_n - s_n)\big).$$

This is bounded away from zero because $\sup_{q\geq d} \operatorname{sinc}(q) < 1$, by continuity and decay at infinity. These cases establish (A.2).

Write $K = \operatorname{Lip}(\gamma')$ and $K_u = \operatorname{Lip}(u)$. Choose $d > 0$ with $Kd \leq c/2$. The estimate

$$\|\gamma(t) - \gamma(s) - (t - s)\gamma'(s)\| \leq \frac{K}{2}|t - s|^2$$

then implies

$$|\langle \gamma(t) - \gamma(s), u(s)\rangle| \geq \frac{3c}{4}|t-s| \qquad (|t-s| \leq d). \tag{A.3}$$

Choose $\rho > 0$ so that

$$\rho < \frac{1}{32}, \qquad 4\rho < \sigma(d), \qquad \rho K_u < \frac{c}{4}. \tag{A.4}$$

For $x \in U$, take a minimizing sequence $\gamma(t_n)$ for the distance from x to the curve. Its late terms satisfy $\|x - \gamma(t_n)\| < 2\rho$, so their pairwise distances are less than 4ρ. By (A.2) their parameters differ by less than d. Fixing one late parameter therefore confines all later ones to a compact real interval. A convergent subsequence attains the minimum. Differentiating $\|x - \gamma(t)\|^2$ at any minimizing parameter gives $(x - \gamma(t)) \perp u(t)$.

For $x, y \in U$ with $\|x - y\| < \rho$, choose any minimizing parameters t, s and write $x = \gamma(t) + e_t$, $y = \gamma(s) + e_s$. The curve points are less than 3ρ apart, so $|t - s| < d$. Orthogonality and (A.3) give

$$\begin{aligned} \|x - y\| &\geq |\langle \gamma(t) - \gamma(s), u(s)\rangle| - |\langle e_t, u(s) - u(t)\rangle| \\ &\geq \left(\frac{3c}{4} - \rho K_u\right)|t - s| \geq \frac{c}{2}|t - s|. \end{aligned}$$

Taking $y = x$ proves uniqueness; the same inequality proves (A.1). □

We next arrange that a residual perpendicular to the curve's tangent is carried to a vector perpendicular to $v(t)$. This will let us change the coefficient along $v(t)$ without cancellation by that residual.

Lemma A.3. *There is a family of orthogonal operators $R_t : H \to H$, Lipschitz in operator norm, such that $R_t u(t) = v(t)$ and R_t is the identity for $t \leq 1$.*

Proof. Suppress t and put $\kappa = \langle u, v\rangle \geq 0$. Define

$$Ah = v\langle u, h\rangle - u\langle v, h\rangle, \qquad R = \mathrm{Id} + A + \frac{A^2}{1+\kappa}.$$

If u, v are independent, then A is skew-adjoint, $A^2 = -(1-\kappa^2)\mathrm{Id}$ on their span, and R restricts there to $\kappa\mathrm{Id} + A$. This restriction is orthogonal and sends u to v, while R is the identity on the orthogonal complement. If $u = v$, then $A = 0$ and $R = \mathrm{Id}$. These exhaust the cases because $\kappa \geq 0$. The formula, the Lipschitz dependence of u, v, and $1 + \kappa \geq 1$ show that $t \mapsto R_t$ is Lipschitz in operator norm, including where $u = v$. □

Proof of Proposition A.1. Fix the tube radius from Lemma A.2 and the rotations from Lemma A.3. Set

$$b(t) = \min\{a(t), -1/8\}, \qquad q(t) = b(t) - a(t) = -\max\{a(t) + 1/8, 0\}.$$

Then $|q(t)| \leq 7/16$ and $\mathrm{Lip}(q) \leq 1/8$. Let $\beta, \chi : [0, \infty) \to [0, 1]$ be the continuous piecewise linear functions specified by

$$\beta(r) = \begin{cases} 1, & r \leq \rho/4, \\ 2 - 4r/\rho, & \rho/4 < r < \rho/2, \\ 0, & r \geq \rho/2, \end{cases} \qquad \chi(r) = \begin{cases} 1, & r \leq \rho/2, \\ 2 - 2r/\rho, & \rho/2 < r < \rho, \\ 0, & r \geq \rho. \end{cases}$$

Define $G : B \to H$ to equal the identity outside U. For $x \in B \cap U$, write $t = t(x)$, $e = x - \gamma(t)$, $r = \|e\|$, and set

$$\begin{aligned} G(x) &= \gamma(t) + \beta(r)q(t)v(t) + \big((1-\chi(r))\mathrm{Id} + \chi(r)R_t\big)e \\ &= x + \beta(r)q(t)v(t) + \chi(r)(R_t - \mathrm{Id})e. \end{aligned} \tag{A.5}$$

First, G is globally Lipschitz on B. For tube points at distance less than ρ, (A.1) controls their parameters, and, with $M = \sup_t \|\gamma'(t)\|$, their residuals satisfy

$$\|e(x) - e(y)\| \le (1 + 16M)\|x - y\|.$$

Moreover, $r(x) = \mathrm{dist}(x, \gamma(\mathbb{R}))$ is 1-Lipschitz. Every factor in the second line of (A.5) is therefore Lipschitz for such pairs with uniform constants; the factors are uniformly bounded, using $\|e\| < \rho$, $|q| \le 7/16$, $\|v\| = 1$ and $\|R_t\| = 1$. The product estimates give one finite Lipschitz bound for all these pairs. Also, on all of B,

$$\|G(x)\| \le 1 + 7/16 + 2\rho < 3/2,$$

so pairs at distance at least ρ are controlled by $3/\rho$. For the remaining pairs, let $x \in U$ and $y \notin U$. Both cutoffs satisfy $\beta(r), \chi(r) \le 2(\rho - r)/\rho$, whence

$$\|G(x) - x\| \le \left(\frac{7}{8\rho} + 4\right)(\rho - r).$$

Since distance to the curve is 1-Lipschitz and $y \notin U$, $\rho - r \le \|x - y\|$. Together with $G(y) = y$, this proves the required bound across the tube boundary. Outside the tube G is the identity.

Next, we claim that

$$\|G(x)\| \ge \rho/4 \qquad (x \in B). \tag{A.6}$$

Outside U, the fact that $0 = \gamma(0)$ gives $\|G(x)\| = \|x\| \ge \rho$. Inside U, if $r \le \rho/2$, then $\chi(r) = 1$ and $R_t e \perp v(t)$, with $\|R_t e\| = r$. For $r \le \rho/4$, the coefficient along $v(t)$ is $b(t) \le -1/8$; for $\rho/4 \le r \le \rho/2$, the perpendicular component has norm at least $\rho/4$. If $\rho/2 \le r < \rho$, then $\beta(r) = 0$ and the last term in the first line of (A.5) has norm at most $r < \rho$. When $t > 1$, we have $a(t) \ge 1/8$, so $\|G(x)\| \ge 1/8 - \rho > \rho/4$. When $t \le 1$, we have $R_t = \mathrm{Id}$ and $e \perp v(t)$, so $\|G(x)\| \ge r \ge \rho/2$. This proves (A.6).

Finally, $G(x) = x$ whenever $\|x\| \ge 1/2$. This holds by definition outside U. Inside U, the reverse triangle inequality gives

$$|a(t)| = \|\gamma(t)\| \ge \|x\| - r > 1/2 - \rho > 5/16.$$

Since $a(t) \le 5/16$ everywhere, this forces $a(t) < -1/8$ and $t \le 1$. Thus $q(t) = 0$ and $R_t = \mathrm{Id}$, so (A.5) again gives $G(x) = x$.

Define

$$f(x) = -\frac{G(x)}{\|G(x)\|}.$$

Normalization on vectors of norm at least $\rho/4$ is Lipschitz with constant at most $8/\rho$, so f is globally Lipschitz by (A.6), and its values have norm one. If $\|x\| < 1/2$, then $\|f(x) - x\| \ge 1 - \|x\| > 1/2$. Otherwise $G(x) = x$, and $\|f(x) - x\| = 1 + \|x\| \ge 3/2$. □

References

[1] Azer Akhmedov. Non-amenability of R. Thompson's group F. arXiv:1310.4395v3, 2021.

[2] Y. Benyamini and Y. Sternfeld. Spheres in infinite-dimensional normed spaces are Lipschitz contractible. *Proceedings of the American Mathematical Society*, 88(3):439–445, July 1983. doi:10.1090/S0002-9939-1983-0699410-7.

[3] Matthew G. Brin and Craig C. Squier. Groups of piecewise linear homeomorphisms of the real line. *Inventiones Mathematicae*, 79(3):485–498, 1985. doi:10.1007/BF01388519.

[4] Kenneth S. Brown and Ross Geoghegan. An infinite-dimensional torsion-free FP_∞ group. *Inventiones Mathematicae*, 77(2):367–381, 1984. doi:10.1007/BF01388451.

[5] J. W. Cannon, W. J. Floyd, and W. R. Parry. Introductory notes on Richard Thompson's groups. *L'Enseignement Mathématique (2)*, 42(3–4):215–256, 1996. Primary article scan.

[6] Mahlon M. Day. Amenable semigroups. *Illinois Journal of Mathematics*, 1(4):509–544, 1957. doi:10.1215/ijm/1255380675.

[7] Erling Følner. On groups with full Banach mean value. *Mathematica Scandinavica*, 3:243–254, 1955. doi:10.7146/math.scand.a-10442.

[8] Victor Guba. Cayley graphs of R. Thompson's group F: New estimates for the density. *Journal of Combinatorial Algebra*, 9(1/2):145–168, 2025. doi:10.4171/JCA/110.

[9] Victor Guba. On zero-measured subsets of Thompson's group F. *Journal of Algebra*, 692:106–122, April 2026. doi:10.1016/j.jalgebra.2025.12.006.

[10] Uffe Haagerup and Kristian Knudsen Olesen. Non-inner amenability of the Thompson groups T and V. *Journal of Functional Analysis*, 272(11):4838–4852, 2017. doi:10.1016/j.jfa.2017.02.003.

[11] Ralph McKenzie and Richard J. Thompson. An elementary construction of unsolvable word problems in group theory. In W. W. Boone, F. B. Cannonito, and R. C. Lyndon, editors, *Word Problems: Decision Problems and the Burnside Problem in Group Theory*, volume 71 of *Studies in Logic and the Foundations of Mathematics*, pages 457–478. North-Holland, Amsterdam, 1973. doi:10.1016/S0049-237X(08)71914-1.

[12] Justin Tatch Moore. A note on Shavgulidze's papers concerning the amenability problem for Thompson's group F. arXiv:1102.0747v2, 2011.

[13] Justin Tatch Moore. Nonassociative Ramsey theory and the amenability of Thompson's group. arXiv:1209.2063v4, 2012. Withdrawn.

[14] Justin Tatch Moore. Amenability and Ramsey theory. *Fundamenta Mathematicae*, 220(3):263–280, 2013. doi:10.4064/fm220-3-6.

[15] Justin Tatch Moore. Fast growth in the Følner function for Thompson's group F. *Groups, Geometry, and Dynamics*, 7(3):633–651, 2013. doi:10.4171/GGD/201.

[16] OpenAI. Nonuniqueness of percolation on nonamenable quasi-transitive graphs. OpenAI Math Release preprint OAI:Nonuniqueness-of-percolation-on-nonamenable-quasi-transitive-graphs-September-24-2026, 2026. Corollary 1.2.

[17] OpenAI. Unitarizability implies amenability for discrete groups. OpenAI Math Release preprint OAI:Unitarizability-Implies-Amenability-for-Countable-Groups-September-23-2026, 2026. Theorem 1.1.

[18] E. T. Shavgulidze. About amenability of subgroups of the group of diffeomorphisms of the interval. arXiv:0906.0107v1, 2009.

[19] John von Neumann. Zur allgemeinen Theorie des Masses. *Fundamenta Mathematicae*, 13:73–116, 1929. doi:10.4064/fm-13-1-73-116.